DAILY PRACTICE PROBLEM (DPP)

Core Concept Booster Series

Subject: Physics | **Topic: Electrostatics** | **Target: NEET / JEE**

⚡ Quick Revision Formula Key

Coulomb's Law: $F = (1 / 4\pi\varepsilon_0) \times (q_1 q_2 / r^2)$

Electric Field Intensity: $E = F / q_0$

Electric Dipole Moment: $p = q \times 2a$ (From -q to +q)

Gauss's Law: $\Phi = \oint E \cdot dA = q_{enclosed} / \varepsilon_0$

Section A: Multiple Choice Questions (MCQs)

Q1. A body can be negatively charged by:

(a) Giving excess of electrons to it

(b) Removing some electrons from it

(c) Giving some protons to it

(d) Removing some neutrons from it

Q2. Two point charges $+3\mu C$ and $+8\mu C$ repel each other with a force of $40\ N$. If a charge of $-5\mu C$ is added to each of them, then the new electrostatic force between them will be:

(a) -10 N (Attractive)

(b) +10 N (Repulsive)

(c) +20 N (Repulsive)

(d) -20 N (Attractive)

Q3. An electric dipole is placed in a uniform electric field. The net translational force acting on the dipole is:

(a) Always zero

(b) Dependent on the orientation of the dipole

(c) Never zero

(d) Dependent on the strength of the electric field

Section B: Short Answer Conceptual Questions

Q4. Define Electric Field Intensity at a point in space. Write its SI unit and dimensional formula.

Q5. State Coulomb's Law in vector notation. Why is the vector representation considered superior to the scalar form?

Q6. Can two different equipotential surfaces intersect each other? Justify your stance with a logical explanation.

Section C: Numerical & Analytical Problems

Q7. Calculate the total number of elementary electrons required to constitute *1 Coulomb* of negative charge. (Take fundamental electronic charge $e = 1.6 \times 10^{-19}\ C$)

Q8. Two isolated point charges $q_1 = 2 \times 10^{-7}\ C$ and $q_2 = 3 \times 10^{-7}\ C$ are placed exactly *30 cm* apart in free air space. Determine the absolute magnitude and specific direction of the electrostatic force acting between them.

💡 High-Scoring Expert Pro-Tips

- **Unit Consistency Check:** Always convert distances from centimeters (*cm*) to meters (*m*) and micro-Coulombs (μC) to Standard Coulombs (C) before starting calculations.
- **Field Mapping Rule:** Remember that electric field lines always originate perpendicularly from positive charge surfaces and terminate on negative charge boundaries.
- **Symmetry in Gauss's Law:** Always choose a Gaussian surface that mirrors the symmetry of the charge distribution (spherical, cylindrical, or planar) to solve flux integrals instantly.